

LESSON 3.6

SOLVING REAL-WORLD PROBLEMS WITH RATIONAL NUMBERS – PART 2



LESSON INTRODUCTION

MA.7.NSO.2.3 Solve real-world problems involving any of the four operations with rational numbers.

LEARNING TARGETS

- I can solve problems with one or two operations by multiplying and/or dividing rational numbers.

BENCHMARK COHERENCE

BUILDING ON

MA.6.NSO.2.3 Solve multi-step, real-world problems involving any of the four operations with positive multi-digit decimals or positive fractions, including mixed numbers.

WORKING TOWARD

MA.8.NSO.1.7 Solve multi-step mathematical and real-world problems involving the order of operations with rational numbers including exponents and radicals.

ESSENTIAL QUESTIONS

- How do you solve multi-step real-world problems using multiplication and division of rational numbers?

LESSON NARRATIVE

Students transition from just using addition and subtraction to solve real-world problems to include using multiplication and division to solve real-world problems. In this lesson, students are presented with a real-world, multi-step problems with rational numbers and collaborate to solve by drawing a diagram and answering questions that break the problem down into simpler steps. Students pull key information from a given problem and break it into simpler steps while they practice applying these skills to solve other problems.

COMPONENT SUMMARY

3.6.1 WARM-UP (~5 MIN.)

Identify a math fail on a thermometer

3.6.3 GUIDED INSTRUCTION (15-20 MIN.)

Using key information and drawing diagrams, break into steps to solve real-world, multi-step problems with rational numbers

3.6.5 WRAP-UP (~5 MIN.)

Solve a real-world, multi-step problem with rational numbers

3.6.2 COLLABORATION (10-15 MIN.)

Draw a diagram to solve real-world problems with rational numbers

3.6.4 YOUR TURN! (10 MIN.)

Solve real-world, multi-step problems with rational numbers

RESOURCES

GLOSSARY

Key Terms:

- N/A

Additional Terms:

- N/A

MATERIALS

- Calculator

ADDITIONAL PREPARATION

- 3.6.1 Warm-Up has students identifying a math fail on a thermometer. Consider projecting or enlarging the thermometer for students who might struggle with reading the scale.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may not recognize that when asked to describe how a value changes after a decrease, a negative number should be used.

THINGS TO CONSIDER

- Students have just finished four lessons on procedural fluency of the four operations with rational numbers, and one lesson on problem solving with rational numbers. Consider utilizing Study Expert videos for scaffolding on the four operations (Lessons 1-4).
- If time is an issue, then consider assigning 3.6.4 Your Turn! for homework.
- Allowing students to use calculators during the lesson will help students focus on the real-world contexts. Consider having no-calculator questions where students turn their calculator over. Teachers should look for and consider opportunities for students to improve their fluency with rational numbers. For example, questions 2-4 in 3.6.3 Guided Instruction could be a no calculator question but question 1 in 3.6.3 Guided Instruction should be a calculator question. Both 3.6.4 Your Turn! and the 3.6.5 Wrap-Up can be done without a calculator.

LITERACY AND LANGUAGE ROUTINES

Three Reads

3.6.2 Collaboration presents students with a lot of text to work through when solving the given question. Consider utilizing “Three Reads” when presenting the question, positioning students to make sense of the context, highlight numbers and quantities, and identify key information in the question before beginning to solve.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.7.1 Real-World Contexts

Students engage in real-world problem solving throughout the lesson. MTR 7 asks students to “Use models and methods to understand, represent and solve problems.” Students utilize ‘Three Reads’ in this lesson to help them make sense of the question as they read for context, to highlight numbers and quantities, and to identify key information, drawing diagrams, and breaking the questions down into simpler steps to solve.

DIFFERENTIATE INSTRUCTION

For text-heavy lessons like Lesson 6, language development and reading comprehension strategies should be included with instruction. Diagrams can provide a visual entry point for students to engage with the content and make connections between the question, representation, and equation to solve. Additional comprehension strategies include the ‘Three Reads’ routine during 3.6.2 Collaboration as well as reading the question out loud, as struggling readers have an entry point into the activity.

3.6.1 WARM-UP: MATH FAIL

DIFFERENTIATE INSTRUCTION

For students who may struggle to read the thermometer with smaller print, consider projecting a larger image of it for students to use while answering the questions.

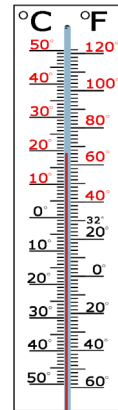


3.6.1 Warm-Up: Math Fail

A picture of a thermometer is shown.

1. Explain the error in the image.

The numbers below zero should have a negative sign. The label for 32° is in the wrong place.



2. Describe how to fix the image so that there is no "math fail."

Add a negative sign to all of the numbers below zero. 32° should be across from 0°.

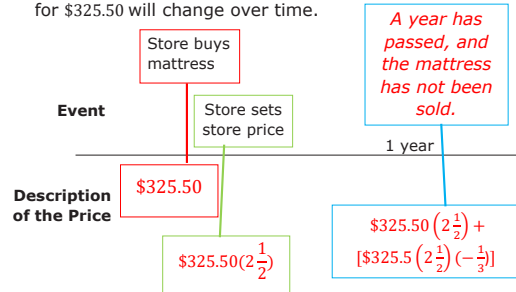


3.6.2 Collaboration: Solving a Problem with Rational Numbers

In order to make a profit, stores sell products for more than what they paid for them. For example, a local furniture store buys a mattress from a manufacturer for a certain price. The store then sets the store price at $2\frac{1}{2}$ times what they paid the manufacturer for it. If the mattress hasn't sold in a year, then the store will reduce its price by $\frac{1}{3}$ of the original store price.

Work with your partner to complete the following based on the information given.

1. Drawing a diagram can help make sense of the problem. Without calculating, complete the timeline to show how the price of a mattress the store purchases for \$325.50 will change over time.



3.6.2 COLLABORATION: SOLVING A PROBLEM WITH RATIONAL NUMBERS

TEACHER MOVES

Three Reads

Consider engaging students in 'Three Reads' to prepare them to make sense of the question.

Begin by reading the question aloud to the whole group after posing the prompt, "What is this situation about?" Students sit with their partner and discuss, sharing with the whole group when finished.

Pose the second prompt, "What are the quantities in this situation?" Have partners read through the second time and bring the whole group together again; list quantities identified on chart paper or whiteboard.

Pose the third prompt, "What mathematical questions can we ask about this situation?" Have partners again read through for a third time, asking them to come up with two questions. Once again, bring the whole group together to share their questions. Consider prompting students to think about the context and identified quantities by asking, "Can this question be answered using the information we've already identified in the problem?"

At this point, partners begin to collaborate to work through and answer the questions.

3.6.2 COLLABORATION: SOLVING A PROBLEM WITH RATIONAL NUMBERS (CONTINUED)

ENRICHMENT

Refer to information recorded in 'Three Reads' routine and the timeline students filled in for question 1:

- What quantity describes how the store sets the price for the mattress?
- Which price represents what the store paid for the mattress?
- How does the price change once the store sets the price for the mattress?
- Which quantity represents the change in the value of the mattress after one year?

QUESTIONING

Encourage students to answer one or two of the identified questions in the 'Three Reads' practice at the beginning of 3.6.2 Collaboration.

2. Explain whether the store's first price will be greater than or less than the price they paid the manufacturer.

Answers will vary. The store's first price will be greater than the price the local furniture store paid the manufacturer because the store marked up the price $2\frac{1}{2}$ times.

3. What is the first price the store will set for the mattress?

\$813.75

4. Explain whether the price after a year will be greater than or less than the store's first price.

Answers will vary. The price after a year will be less than the store's first price because the store reduces the price by $\frac{1}{3}$ of the original store price to increase the chance of selling the mattress.

5. What is the sale price of the mattress after one year?

\$542.50

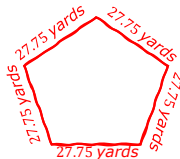
6. If the store sells the mattress after one year, explain if the store will make or lose money on the mattress.

Answers will vary. The store will make a profit from the original purchase of the mattress for \$325.50 after the original increase and reduction in price after a year.



3.6.3 Guided Instruction: Solving Real-World Problems with Rational Numbers

1. Ramon paid \$18,360.00 for a new car. By the end of the first year, the value of the car changed by $-\frac{3}{14}$ of its original value. By the end of the second year, the value of the car changed by $-\frac{2}{9}$ of its value at the end of the first year.
 - a. Is the value of the car increasing or decreasing during these two years?
The value of the car is decreasing during the two years after purchase.
 - b. By what amount did the value of the car change at the end of the first year?
-\$3,934.29
 - c. What is the value of Ramon's car at the end of the second year?
\$11,220
2. A farmer is building a fence around a space on his property. The space has five equal sides. The farmer determined that he should buy 138.75 yards (yd) of fencing. The fence costs \$133.21 per yd.
 - a. Draw and label a diagram to make sense of the problem.



- b. How much fencing is used for each side?
 $138.75 \div 5 = 27.75$ yards

3.6.3 GUIDED INSTRUCTION: SOLVING REAL-WORLD PROBLEMS WITH RATIONAL NUMBERS

TEACHER MOVES

Provide students with some background information on depreciation. Most goods have a depreciation. Some goods may depreciate and then after time their value increases because they become antiques. Explain to students that the $-\frac{3}{14}$ means that this is the amount of the value that the car loses during its first year. Another way of stating that is that at the end of the first year the car is $\frac{11}{14}$ of its original value.

Students may not identify the value of the car after the first year decreased, so the change in value is represented by a negative number. Be flexible here because students may intuitively think in terms of subtraction. Allow a student to explain their answer. Consider asking which students wrote a positive value and which wrote a negative value. Then, choose one student from each to explain their reasoning. If students state that the car is depreciating \$3,934.29 then the negative is inferred.

DIFFERENTIATE INSTRUCTION

Consider providing diagrams for students to label throughout this section. Connecting the visual representation to the question will provide an additional entry point for learners to reason about the steps to take to solve each question.

3.6.3 GUIDED INSTRUCTION: SOLVING REAL-WORLD PROBLEMS WITH RATIONAL NUMBERS (CONTINUED)

TEACHER MOVES

Students may struggle to determine the answers for each question. If students are struggling to get started, ask scaffolding questions to guide students through the steps towards finding their answer.

- How can you use a rate to determine distance? If a car drove 50 miles per hour how far would it travel in 3 hours?
- What is the rate in question 3 (or 4)?
- Which operation can you use to find the time? (If a car drove 150 miles at a rate of 50 mph, how would you determine the time?)
- How can you represent this question with a diagram or table?

- c. What is the cost for fencing on one side of the space?
 $27.75 \times \$133.21 = \$3,696.58$
3. A water well drilling rig dug to a depth of -147.5 feet (ft). The rig drills $-2\frac{1}{2}$ ft in one hour.
 - a. How many hours has the drill been running?
 $-147.5 \div (-2.5) = 59 \text{ hours}$
 - b. At this rate, how many hours will it take until the drill reaches a depth of -250 ft?
 $-250 \div (-2.5) = 100 \text{ hours}$
4. A marine biologist was doing research in a one-person submarine. The submarine started to ascend (go up) from a position of $-130\frac{1}{2}$ feet. The submarine surfaces (goes up) at a rate of 23.3 feet per minute. How long, in minutes, will it take the submarine to reach the surface?
 $130.5 \div 23.3 \approx 5.6 \text{ minutes}$

**3.6.4 Your Turn!**

1. Delaney wants to sell her bicycle. The original price of the bike was \$150. The value of the bike has changed by $-\frac{2}{5}$ of the original price. What is the current value of the bike?

\$90

2. At the beginning of the day on Monday, Stephon had \$15.78 in his lunch account. At the end of the day on Friday, he had \$1.53 in his lunch account. If Stephon purchased the same lunch each day on Monday through Friday, describe how the account changed each day.

*The account decreased by \$2.85 per day.***3.6.4 YOUR TURN!****TEACHER MOVES**

Encourage students who may be struggling with question 2 to refer to the diagram they used in 3.6.2 Collaboration question 1.

**3.6.5 Wrap-Up: Ticket Out the Door**

1. If the temperature has been changing at a rate of -3.64°F every hour for $3\frac{1}{2}$ hours, how much has the temperature changed over that time?

 -12.74°F **3.6.5 WRAP-UP: TICKET OUT THE DOOR****DIFFERENTIATE INSTRUCTION**

Encourage students to create a diagram of the scenario to represent how to answer the question in order to provide a visual entry point.